# Shourya Upadhyaya

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## EXPERIENCE

**Moss X** | mossx.in | GitHub — Apr. 2025 – Oct. 2025

*Lead Full-Stack Developer* — *Remote*

- Architected a modular monolithic e-commerce backend serving **100+ registered users** with **Node.js, Express.js, MongoDB, and React**, building multi-stage aggregation pipelines for sales analytics, catalog filtering, and guest-to-cloud cart merging.
- Engineered a **Redis** cache-aside layer with automated invalidation on catalog updates, cutting product listing API latency by **90% (1.5 s to 150 ms)** and sustaining **sub-50 ms** homepage load times.
- Integrated **Razorpay** checkout flows with cryptographic **HMAC webhook verification**, custom **idempotency keys** (**0 duplicate charges** across **20+ live orders**), and dual-token **JWT + HttpOnly** refresh authentication.
- Led a team of **3 developers** across frontend architecture, pull-request reviews, and feature delivery while containerizing services with **Docker** for production deployments on **Render**.

**Nepbyte Technology** — Sept. 2024 – Feb. 2025

*Frontend Engineering Intern* — *Remote*

- Accelerated **Largest Contentful Paint (LCP) by 49% (3.5 s to 1.8 s)** and boosted URLs passing **Core Web Vitals from 45% to 85%** through route-based code splitting, dynamic lazy loading, and semantic HTML.
- Built modular **React.js** interfaces using **Tailwind CSS** and **Zustand** state stores, implementing **Axios** request/response interceptors for automated token refresh and standardized REST API error handling.

## PROJECTS

**OpenPumTa** | *Next.js, Express.js, TypeScript, PostgreSQL, Prisma, Redis, Docker* | openpumta.com | GitHub — Jan. 2025 – Present

- Scaled a full-stack productivity workspace to **300+ registered users** and **10,000+ visits**, developing **30+ Zod-validated REST APIs** and **10 Prisma/PostgreSQL models** for habits, tasks, and focus sessions.
- Reduced **PostgreSQL read load by 80%** via **Redis** analytics caching and engineered a persistent **WebSocket** state-synchronization engine for **sub-50 ms** cross-device Pomodoro timer accuracy.
- Deployed the multi-container **Docker Compose** stack on a self-hosted **Linux VPS** behind **Nginx** (SSL termination), instrumenting **Prometheus** scrapers and **Grafana** dashboards to monitor p95 latency, active sockets, and error rates.
- Integrated **Groq LLM** pipelines for automated daily burnout-prevention reviews, backed by centralized async error handling and automated integration tests.

**VoiceShield AI** | *Node.js, TypeScript, Python, FastAPI, PyTorch, Whisper, WebSockets* | GitHub — Smart India Hackathon Finalist

- Engineered a real-time **Node.js/TypeScript** media gateway ingesting **48 kHz mono PCM16** audio over **WebSockets**, reassembling arbitrary network frames into ordered **3-second chunks** verified by a strict byte-conservation invariant (**50+ unit tests**).
- Decoupled **PostgreSQL** persistence off the hot audio path via asynchronous write queues while routing binary chunks over persistent WebSockets to a **FastAPI/PyTorch** inference worker and streaming live alerts to **Next.js** via **Socket.IO**.
- Fused **OpenAI Whisper** transcripts with acoustic deepfake signals (**RawNet2, WavLM, ONNX**) inside a scam-risk agent, achieving a **1.1 s median CPU inference latency** per 3-second audio window (**2.7x faster than real-time**).

**Cook Off 10.0 Portal** | *Next.js, TypeScript, CodeMirror 6, Zustand, MongoDB* | GitHub — Sept. 2025 – Feb. 2026

- Architected a competitive programming IDE on **Next.js** and **CodeMirror 6** for **CodeChef-VIT**, sustaining **40% peak concurrent load across 1,000+ contest users** with **zero dropped submissions** during final-minute traffic spikes.
- Designed fault-tolerant **triple-layer state persistence** (**Zustand, LocalStorage, MongoDB**), automated timer-expiry submission, and **Axios** auth interceptors delivering sub-second multi-language compiler feedback.

**CRMS & GoLakeScanner** | *PostgreSQL 16, Go (pgx/v5), TypeScript, Express.js, Docker* | CRMS | GoScanner — 2026

- Engineered a zero-ORM cloud resource plane in **PostgreSQL 16** enforcing **Row-Level Security (RLS)** and `SELECT FOR UPDATE` quota locking; cut **1M+ row** query latency by **99.5% (42 ms to 0.2 ms)** via **BRIN/GIN** indexing and streamed `pg_notify` alerts over **SSE**.
- Built an incremental **Go** data-lake indexer using **SHA-256** content addressing and a generic bounded goroutine worker pool (`pool.Run[In, Out]`) with **30 s per-file context timeouts** and **pgx/v5** cataloging.

## TECHNICAL SKILLS

**Languages:** TypeScript, JavaScript, Python, Go (Golang), C++, SQL (PostgreSQL)
**Backend & Streaming:** Node.js, Express.js, FastAPI, WebSockets, Socket.IO, SSE, REST APIs, Prisma ORM, pgx/v5
**Frontend:** React.js, Next.js, Tailwind CSS, Zustand, TanStack Query, CodeMirror 6
**Databases & Caching:** PostgreSQL (RLS, Partitioning, PL/pgSQL, BRIN/GIN), MongoDB (Aggregation Pipelines), Redis
**AI / ML & Audio:** PyTorch, OpenAI Whisper, ONNX Runtime, SpeechBrain (ECAPA-TDNN), Groq & Gemini APIs
**DevOps & Tools:** Docker, Docker Compose, Nginx, Linux VPS, Prometheus, Grafana, Git, Vitest, Playwright

## EDUCATION & ACHIEVEMENTS

**Vellore Institute of Technology (VIT Vellore)** — Aug. 2024 – May 2028

*B.Tech. in Computer Science and Engineering | Coursework: DSA, OOP, DBMS, Computer Networks, OS* — *Vellore, India*

- **Smart India Hackathon (SIH) College Finalist:** Selected among top teams at VIT Vellore for building VoiceShield AI's real-time audio deepfake and scam-detection pipeline.
- **Technical Engineering Writer (Medium):** Published system-design deep dives on *Linux VPS Docker Consolidation*, *Drift-Free WebSocket Pomodoro Timers*, and *Express.js Async Architecture*.